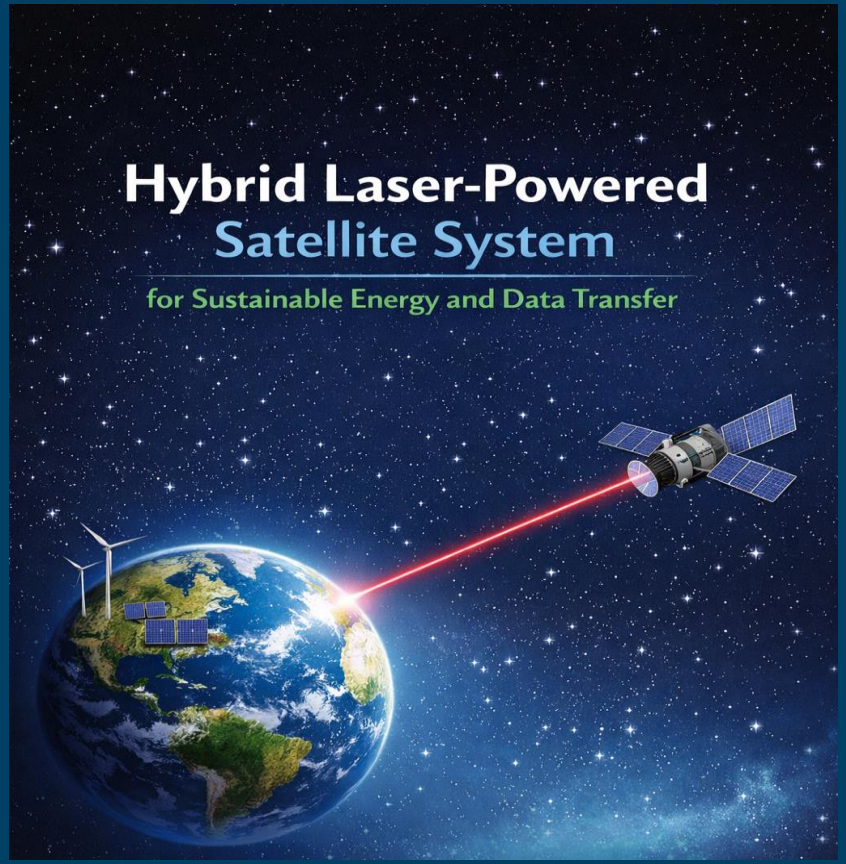
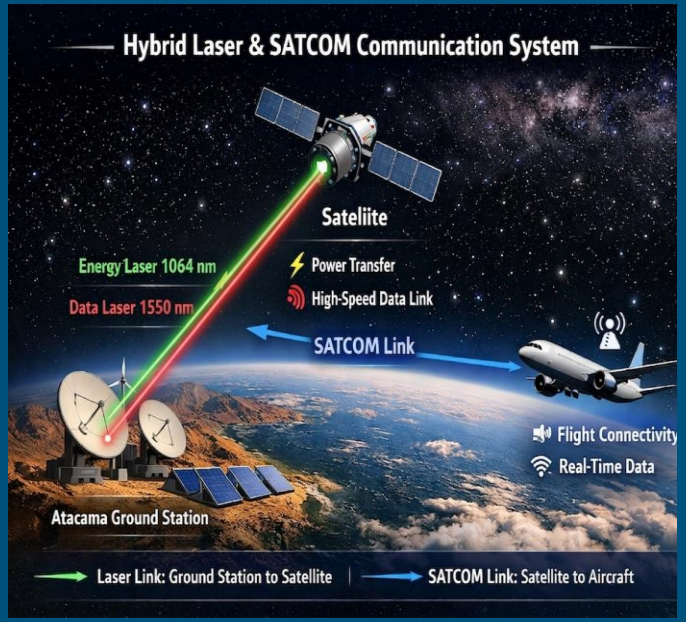
Hybrid Satellite Energy & Laser Communication System Architecture



AV.CREW team from Amman Arab University, Jordan

General overview

The project focuses on establishing a hybrid communication system in the Atacama Desert, relying on laser ground stations linked to a satellite to communicate with an aircraft over the ocean. The goal is to use lasers to transmit data at higher speeds and with greater efficiency than radio waves, with a SATCOM channel serving as a backup to ensure continuity. This model reflects the latest trends from NASA, ESA, and IEEE in the development of secure and efficient space networks, and demonstrates how optical communications can support both aviation and space operations in a harsh environment such as the Atacama.



Materials & Equipment .

Actual Materials and Equipment Used in the Project

.1Atacama Ground Station

-Solar panels for power generation .

-Ground-based SATCOM antennas .

-Nd:YAG laser unit with a wavelength of 1064 nm (for power transmission) .

-Diode/Fiber laser unit with a wavelength of 1550 nm (for data transmission) .

-Optical detectors/receivers .

-Precision stabilization and alignment systems for the laser beam .

.2Satellite Relay

-Dual laser receivers (for power and data) .

-Relay module .

-SATCOM antenna for redistributing data to the aircraft .

-Optical stabilization and alignment systems .

.3Aircraft

-SATCOM antenna for receiving data from the satellite .

-Real-Time Data Receiver .

-Flight data management systems .

Stages of the Device's Operation (Slides Plan)

◆Phase – noitareneG ataD dna rewoP :1Ground Station

-Location: Ground station in the Atacama Desert (dry climate, clear skies) .

-Equipment: Solar panels, Nd:YAG laser (1064 nm), Diode/Fiber laser (1550 nm) .

-Function: Generate electricity from the sun →convert it into an energy beam + transmit data via laser .

-Control Algorithm: A laser alignment system that adjusts the transmission angle based on the satellite's position.

Phase – etilletaS eht ot noissimsnarT :2Laser Link

-Location: A laser beam directed from Atacama toward the satellite .

-Equipment: Optical alignment systems, laser receivers on the satellite .

-Function: Transmission of power and data via two separate beams .

-Control algorithm: Tracking the satellite using GPS data and optical sensors to ensure beam stability.

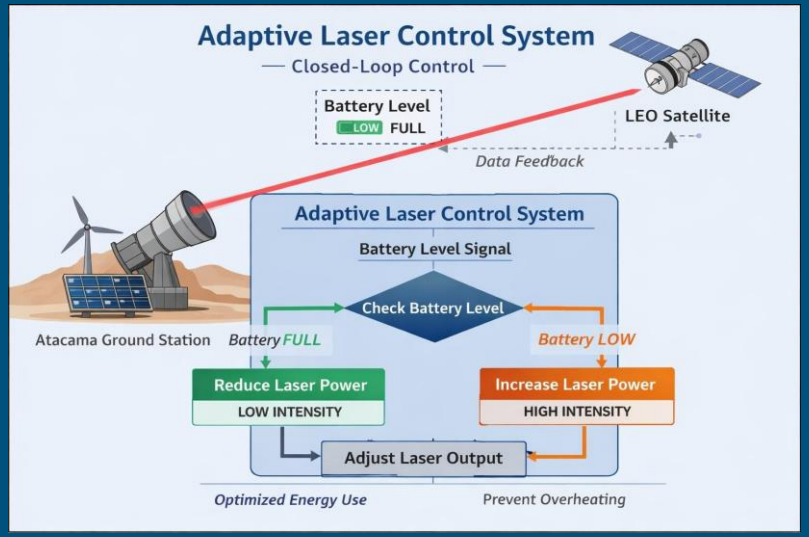
Phase – gnissecorP dna noitpeceR :3Satellite Relay

-Location: Satellite in orbit .

-Equipment: Optical receivers, relay module .

-Function: Receiving power and data via laser, processing the signal, and preparing it for redistribution via SATCOM .

-Control Algorithm: Balance between laser and SATCOM signals, select the optimal path based on weather conditions.



Note on the algorithm :
A closed-loop control algorithm is used to adjust the amount of received energy in real time. Sensors measure the intensity of the laser beam; the system compares this to the desired reference value and then continuously adjusts the laser power or beam angle to ensure a consistent and safe delivery of the appropriate energy, even as weather conditions change or the beam is scattered.

Phase – noitubirtsideR :4SATCOM Link

-Location: From the satellite to the aircraft over the ocean .

-Tools: SATCOM antenna on the satellite .

-Function: Secure and continuous data transmission even if the laser is affected by weather conditions .

-Control Algorithm: Automatic switching protocol between the laser and SATCOM to ensure real-time communication

Phase ytivitcennoC tfarcriA :5

-Location: The aircraft in flight .

-Equipment: SATCOM antenna, flight data receiver .

-Function: Receive real-time data and display it in the cockpit .

-Control algorithm: Manage data and distribute it to flight systems in real time.

Power Consumption Modes

— Energy Usage in Different Operating States —

ACTIVE

— HIGH POWER —

Laser: High Intensity

Laser: High Intensity

Energy Systems: Full Output

Cooling: Maximum

IDLE

— LOW POWER —

Laser: Low Intensity

Control System: Monitoring Only

Energy Systems: Partial Output

SLEEP

— MINIMAL POWER —

Laser: **Off** (Sleep Mode)

Control System: Standby Mode

Energy Systems: *Low or Off*

Cooling: Powered Down

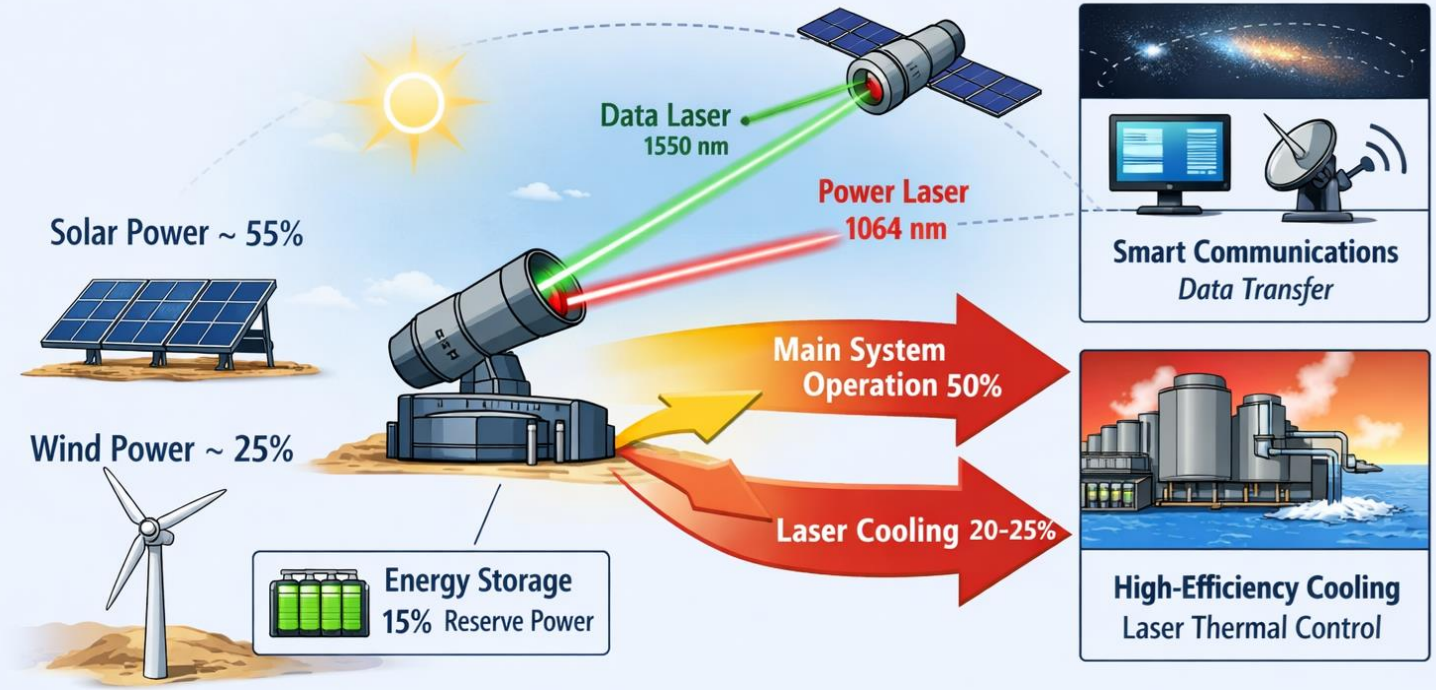
Adjust Laser Output

Optimized Energy Use

Extend Equipment Life

Energy Distribution for Satellite System

Solar & Wind Power in the Atacama Desert



Renewable Energy

50% Main System Operation + 20–25% Laser Cooling + 15% Battery Storage

Advantages and disadvantages

Advantages (in terms of sustainability)

-Use of solar energy: Reduces reliance on fossil fuels and promotes clean energy .

-High data transmission efficiency: Reduces the need for massive ground-based infrastructure .

-Reduced carbon emissions: Because the system relies on lasers and solar energy instead of traditional stations .

-Extended system lifespan: Thanks to closed-loop control technologies that maintain power stability .

-Scalability: Ground stations can be deployed in remote areas without significant environmental impact .

Disadvantages (in terms of sustainability)

-Susceptibility to weather conditions: Dust or clouds may reduce laser efficiency and affect power stability .

-High initial cost: Manufacturing high-power lasers and optical receivers requires significant investment .

-Complex maintenance: Precision optical systems require constant monitoring to maintain efficiency .

-Location dependence: The system's success is tied to regions with dry, clear climates such as the Atacama Desert, limiting global deployment .

-Consumption of rare materials: Manufacturing certain laser components requires rare metals, which may pose environmental challenges .

Key Components and Their Rationale:

-Laser Source ：

Selected because it provides a high-intensity beam that can be precisely controlled, ensuring efficient and sustainable transmission of energy and data ．

-Optical Receivers ：

Essential for converting the laser beam into processable electrical signals, with high sensitivity to minimize loss ．

-Relay Module ：

Ensures signal stability and redistribution via SATCOM, enhancing reliability and reducing the need for a massive ground infrastructur

-SATCOM Antenna ：

Provides a traditional backup communication channel and serves as a contingency solution to ensure continuity in challenging weather conditions ．

-Closed-Loop Control Algorithm ：

Responsible for adjusting the amount of received power in real time, maintaining efficiency and preventing waste or interruptions ．

Future Applications in Aviation

Potential Uses:

-Long-range aircraft communications: Connecting aircraft to a high-speed data network via lasers and SATCOM, even over oceans and remote areas .

-Powering electric or hybrid aircraft: Transmitting a portion of the power via laser to support sustainable flight systems .

-Improving aviation safety: Providing accurate, real-time weather and navigation data, thereby reducing accidents caused by poor communication .

-Supporting unmanned aerial vehicles (UAVs): Enabling the operation of large fleets of UAVs via stable and sustainable communications .

-Reducing traditional ground infrastructure: Reducing reliance on massive radio stations, thereby helping to minimize environmental impact .

Contributors to this work and idea

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Thank you